Question

The metal **M** can react with gas **A** under heating to form a binary compound **B**, which is soluble in both acids and bases. When **B** is dissolved in dilute hydrochloric acid, no gas is released. Dissolving **B** in sodium hydroxide solution and heating yields a solution of **C** and gas **A**, which turns moist red litmus paper blue. Passing hydrogen sulfide gas into the colorless solution **C** produces a white precipitate **D**. **D** is insoluble in water but soluble in hydrochloric acid. Gradually adding dilute hydrochloric acid to solution **C** initially forms a white precipitate **E**, which is soluble in ammonia water. If excess hydrochloric acid is added, the precipitate dissolves to form a colorless solution **F**. **M** can also react with iodoethane to yield a non-iodine-containing compound **G**. When exposed to air, **G** is oxidized by oxygen to form compound **H**, which contains no peroxide bonds. Dissolving 1 mole of **H** in excess sodium hydroxide solution still yields 1 mole of **C** and releases 1 mole of ethane.

Based on the above information, analyze the chemical formulas of each substance and select the correct option.

A. All other options are incorrect

B. Gas **A** is an inert gas.

C. Metal **M** is commonly used in welding

D. The molecular weight of compound **B** is approximately 84

E. Substance **C** contains three elements.

F. Compound **D** can be used as phosphors and pigments.

G. Precipitate **E** has a molecular weight of approximately 146.

H. Substance **F** is not prone to deliquescence.

I. After heating, compound **G** decomposes to form a compound with a molecular weight of 76.

J. Compound **H** contains an anion.

Answer

Correct Answer: F

Detailed Explanation

From the fact that **A** can turn litmus blue, it is known to be ammonia (NH_3).

CHECKPOINT 1 PTS

Gas **A** is ammonia

Next, since metal **M** can react with ammonia, it is known to be reactive; since its sulfide precipitate can be transformed by acid, its sulfide solubility is not very poor; since it can form solutions in both acidic and alkaline environments, it is amphoteric; and since it can react with iodoethane to form a compound without iodine, it can form organometallic compounds.

Therefore, from the aforementioned properties, the one that matches these characteristics is selected, and by further considering the color of its sulfide, Sn is excluded, confirming that metal **M** is zinc (Zn).

CHECKPOINT 5 PTS

Metal **M** is zinc

Thus, the binary compound **B** is formed by the reaction of ammonia and zinc metal, which should be Zn_3N_2.

CHECKPOINT 1 PTS

Compound **B** is Zn_3N_2

Compound **C** is the product dissolved in alkali, so it should be $Na_2Zn(OH)_4$.

CHECKPOINT 1 PTS

Compound **C** is $Na_2Zn(OH)_4$

Compound **D** is the sulfide precipitate formed, which is ZnS, and can be used in pigments and fluorescence.

CHECKPOINT 0.5 PTS

Compound **D** is ZnS

Compound **E** is $Zn(OH)_2$.

CHECKPOINT 0.5 PTS

Compound **E** is $Zn(OH)_2$

Compound **F** is the chloride, which should be $ZnCl_2$, a substance that is highly hygroscopic.

CHECKPOINT 0.5 PTS

Compound **F** is $ZnCl_2$

Compound **G** is the organometallic compound formed by the reaction with iodoethane. Since it does not contain iodine, it is identified as $Zn(C_2H_5)_2$. Heating it generates ethylene with a molecular weight of 28.

CHECKPOINT 1 PTS

Compound **G** is $Zn(C_2H_5)_2$

Compound **H** is the oxidation product of **G**. It dissolves in alkali to produce only one equivalent of ethane and contains no peroxide bond, so it is identified as $Zn(C_2H_5)(OC_2H_5)$.

CHECKPOINT 1.5 PTS

Compound **H** is $Zn(C_2H_5)(OC_2H_5)$

After deducing all the chemical formulas, it can be concluded based on the properties of the compounds that only option F is correct.

